Task Management System

Python To-Do App

PYTHON

JSON

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Problem Description

The Problem

- Many tasks have different deadlines
- Important work can be forgotten
- Students need clear progress tracking

Our Solution

- Store tasks in a dictionary by ID and save them as a JSON list
- Show priority and deadline clearly
- Detect overdue work automatically

Expected Result

- Better organization
- Less missed deadlines
- Simple statistics for progress

A To-Do App is small, but it demonstrates real programming concepts: data structures, validation, sorting, dates, and statistics.

Core Requirements

Add

- Create a new task with title, description, priority, deadline, status

Edit

- Update task information by ID

Delete

- Remove unnecessary tasks safely

Complete

- Mark task as completed

Sort

- Order by priority and deadline

Analyze

- Show overdue tasks and completion statistics

System Design

task object: JSON

```
Each task contains:
• ID
• Title
• Description
• Priority
• Deadline
• Status

The program uses this data to
sort tasks, find overdue tasks,
and calculate statistics.
```

1. Input
- Read JSON task list

2. Validate
- Check priority, status, date

3. Process
- Add/edit/delete/sort tasks

4. Output
- Show sorted tasks, overdue list, stats

Program Demo

```
C:\Users\Admin\Desktop\final-pro\.venv\Scripts\python.exe C:\Users\Admin\Desktop\final-pro\task_management

===== TASK MANAGEMENT SYSTEM =====
1. Show all tasks sorted by priority and deadline
2. Add task
3. Edit task
4. Delete task
5. Mark task as completed
6. Show overdue tasks
7. Show completion statistics
8. Search tasks
9. Show upcoming tasks using generator
0. Save and exit
Choose an option: 1
ID: 1 | Finish project report | Priority: high | Deadline: 2026-05-10 | Status: in_progress | Type: work
ID: 2 | Prepare code defense | Priority: high | Deadline: 2026-05-22 | Status: pending | Type: general
ID: 3 | Buy groceries | Priority: low | Deadline: 2026-05-25 | Status: pending | Type: personal
```

Python Logic

python logic

```
How the Program Works

1. Read tasks from JSON file
2. Check priority, status, and
deadline
3. Sort tasks by priority and
date
4. Find overdue tasks
5. Count completed tasks
6. Show final results
```

Priority Ranking

- High tasks appear first, then medium, then low

Deadline Check

- Date comparison helps detect overdue tasks

Reusable Functions

- Cleaner code and easier debugging

Key Decisions

Complex Parts

- Deadline format validation: YYYY-MM-DD
- Status and priority restrictions
- Sorting by priority and deadline together

Interesting Solutions

- JSON objects represent tasks clearly
- Python dictionaries store task data
- Date comparison detects overdue tasks

Optimizations

- Priority ranking avoids messy conditions
- Reusable functions reduce repeated code
- Simple structure makes future upgrades easier

Team Contribution

Farkhod Sapayev

- Main program logic
- Add / edit / delete tasks
- Status updates and task operations

Nurassyl Serik

- JSON data handling
- Sorting by priority and deadline
- Overdue detection and statistics

Conclusion

Our Task Management System is a modern Python project that organizes tasks, manages deadlines, and tracks completion progress.Through this project, we gained hands-on experience with data structures, JSON handling, input validation, and algorithm design in Python.

Thank You!